MIL-V-8057F
AMENDMENT 3
1 1 NOV 1982
Superseding
AMENDMENT 2
29 April 1980

MILITARY SPECIFICATION

VALVE, OXYGEN, PRESSURE BUILDUP
AND VENT

This amendment forms a part of Military Specification MIL-V-8057F, dated 25 May 1973, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

*2.1: Under "SPECIFICATIONS, Federal", add the following:

"PPP-B-636 - Box, Shipping, Fiberboard
PPP-T-60 - Tape, Packaging, Waterproof
PPP-T-76 - Tape, Packaging, Paper (For Carton Sealing)"

Under "SPECIFICATIONS, Military", delete "MIL-V-25961" and its title, and change the title of MIL-P-116 to read "Preservation, Methods of". Change "MIL-D-1000" to read "DOD-D-1000".

Add the boxed paragraph to the bottom of the page.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

PAGE 2

Under "STANDARDS, Military", change title of MIL-STD-143 to read "Standards and Specifications, Order of Precedence for the Selection of". Change title of MIL-STD-889 to read "Dissimilar Metals".

Under "DRAWINGS", change title of AND10053 to read "Boss, Pipe-Thread, Internal Standard Dimensions for (Inactive For Design Except For Oxygen Systems)". Add the following drawing:

"AF Drawing 8148650 - Cleaning, Inspection and Packaging of Breathing Oxygen Equipment"

3.1: Delete paragraph entirely.

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*3.11.9: Delete "60 pound-inches" and substitute "30 inch-pounds".

FSC 1660

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4.2: Delete "a. Qualification Inspection----(4.3)". Renumber "b" to "a" and "c" to "b".

4.3: Delete paragraph entirely.

4.3.1: Delete paragraph and substitute:

"4.3.1 First article test report. Three copies of the first article test report shall be prepared in accordance with MIL-STD-831 and forwarded to the cognizant engineering activity for approval."

4.3.2: Delete paragraph entirely.

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4.5.1: Add "d. Operating torque (4.7.9)."

4.5.2: Add "f. Extreme low-temperature operation (4.7.10)."

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4.6.2: Delete paragraph and substitute:

"4.6.2 Test fluid. The fluid used in testing the valves shall be either oxygen conforming to MIL-O-27210; air conforming to BB-A-1034, Grade A or C; or nitrogen conforming to BB-N-411, Class 1, Grade A."

4.7.4: Delete the last sentence and substitute, "While maintained at this temperature, they shall be subjected to the vent-port leakage, gas-port leakage, and operating torque tests as specified in 4.7.2, 4.7.3, and 4.7.9, respectively."

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4.7.8: Add the following sentence between the fifth and sixth sentences: "The tests shall be accomplished on the valves while they are in each of three perpendicular positions. The flow rates shall be within the limits specified, in each of the positions."

*4.7.9: Delete paragraph and substitute:

"4.7.9 Operating torque. The valve shall be mounted in a suitable fixture so that a torque will be applied in the direction of travel of the selector handle. The gas port shall be plugged and gaseous pressures of 70 and 300 psig shall be applied to the build-up port while the valve is set in the build-up position. The torque required to set the valve from the build-up position to the vent position at each pressure shall not exceed 30 inch-pounds."

*4.7.10: Delete paragraph and substitute:

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"4.7.10 Extreme low temperature operation. The valve shall be connected to a 300 psi working pressure LOX converter and mounted with the centerline of the stem end oriented downward at least 60 degrees below the horizontal plane in an assimilated aircraft manner. With the valve in the vent position, the converter shall be filled with liquid oxygen at 20 to 40 psi and allowed to run over for one minute. There shall be no evidence of leakage around the valve stem or through any joint in the body during this test. The source will be disconnected from the converter and the valve immediately placed in the build-up position. The operating torque shall not exceed 60 inch-pounds. This operation shall completely cut off the flow of liquid oxygen and there shall be no evidence of leakage of liquid. Leakage through the valve shall be measured at the vent port as the converter pressure builds up throughout the entire range from 0 to 300 psi. If needed, valve rotation to drain any trapped residual liquid causing false vent port leakage is permitted or the test may be repeated in a position that permits valve vent drainage. Following the test, each valve shall be subjected to the vent-port and gas-port leakage tests as specified in 4.7.2 and 4.7.3 respectively."

4.8: Delete the word "Qualification" in the title and last sentence.

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4.8.1: Delete "Qualification" in the first sentence.

*5: Delete section entirely and substitute the following:

"5. PACKAGING

5.1 Preservation-packaging. Preservation shall be Level A.

5.1.1 Level A.

5.1.1.1 Cleaning. Valves shall be cleaned in accordance with AF Drawing 8148650.

5.1.1.2 Drying. Valve shall be dried in accordance with AF Drawing 8148650.

5.1.1.3 Preservation application. Preservatives shall be strictly prohibited in accordance with AF Drawing 8148650.

5.1.1.4 Unit packaging. Unless otherwise specified (see 6.2), valves shall be individually packaged in accordance with MIL-P-116, submethod 1A-8 insuring compliance with the applicable requirements of that specification.

5.2 Packing. Packing shall be Level B or C as specified (see 6.2).

5.2.1 Level B. The packaged valves shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional. Special requirements, in lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, two inches

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minimum width, conforming to PPP-T-60, class I or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using non-metallic or tape banding only.

5.2.2 Level C. The packaged valves shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional. Special requirements, closures shall be in accordance with the appendix thereto.

5.3 Marking. Nomenclature tags, marked in accordance with MIL-STD-129 shall be securely attached to the unit package, showing title, and part number without breaking the seals. In addition, each small unit package shall have a 2" x 6-1/2" or 3" x 10-1/2" label displayed on at least two sides of the container. The color of the labels will be a conspicuous green.

LIFE SUPPORT ITEM
All oil, grease, shop residue or other contaminants have been removed.
Do not open until ready for use."

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*6.3: Delete paragraph and substitute the following:

"6.3 First article. When a first article is required, the item will be inspected and tested in accordance with Section 4 of this document. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article."

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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